

HAMZA KHAN

Data Engineering · Backend Development — Python · SQL · ETL · Django · PostgreSQL

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PROFILE

Python developer with four years of freelance delivery experience building data pipelines and the backend systems that feed them — ETL into PostgreSQL, an analytics engine on DuckDB and FastAPI, Django applications and REST APIs, and reporting in Power BI. Ten systems shipped for clients in India and Germany, each taken solo from requirements through deployment and maintenance, which has meant real servers, real users and debugging in production. B.E. in Computer Science and Engineering (Data Science), University of Mumbai, CGPI 8.19/10. Smart India Hackathon 2025 Grand Finalist as team lead. Seeking an entry-level Data Engineer, Analytics Engineer or Backend Developer role.

TECHNICAL SKILLS

Data Engineering ETL/ELT pipeline design, schema auto-detection, column mapping, data cleaning & validation, de-duplication, incremental loads, data-quality scoring, PII masking (SHA-256), data lineage

Data & Analytics pandas, NumPy, SQLAlchemy, analytical SQL (CTEs, window functions), scikit-learn, XGBoost, Prophet, SHAP, RFM & cohort segmentation, anomaly detection, Power BI, Jupyter

Languages Python, SQL, JavaScript, TypeScript, Bash

Databases PostgreSQL, MySQL, SQLite, DuckDB, Elasticsearch

Backend & APIs Django, Django REST Framework, Django Channels (WebSockets), FastAPI, Flask, REST API design, Swagger/OpenAPI, Postman

Cloud & DevOps Docker, Git, GitHub Actions (CI), Azure, AWS EC2, DigitalOcean, Vercel, Render, Linux, pytest, Ruff

Frontend React, Next.js, Vite, Tailwind CSS, shadcn/ui, TanStack Query, Tauri (desktop)

PROFESSIONAL EXPERIENCE

Founding Developer — NOVEM (Independent Product) — Self-Employed

Dec 2025 – Present

Mumbai, India · Remote · Python, FastAPI, DuckDB, pandas, Tauri (Rust), React

- **Built the full ingestion pipeline** for a local-first e-commerce intelligence platform: schema auto-detection, column mapping, type coercion, cleaning, de-duplication, SHA-256 PII masking and data-quality scoring before any record is merged into the analytical store.
- **Designed the analytical layer** on DuckDB with SQLite for settings, import history and encrypted credentials; sources include CSV, Excel, Google Sheets, Shopify and PostgreSQL.
- **Delivered a FastAPI engine of 23 routers and ~170 endpoints** serving RFM segmentation, churn risk and CLV (BG/NBD + Gamma-Gamma), cohort retention, market-basket rules, Prophet forecasting, anomaly detection with SHAP explanations, and a RAG copilot on a local LLM.
- **Enforced quality with CI on every push** — Ruff, pytest across Python 3.11 and 3.12, tsc --noEmit, ESLint at zero warnings and Vitest — with releases frozen via PyInstaller inside an NSIS installer.

Freelance Full-Stack Developer — Brandenbed Living Spaces UG, Berlin, Germany

Sep 2025 – Nov 2025

Remote · Django, PostgreSQL, Next.js, Docker, Azure

- Built the company's production website for serviced-apartment operations across European markets, achieving a Google PageSpeed Insights score of 95+ on desktop.
- Developed an internal CRM with role-based access controlling property management and client communication workflows for the operations team.
- Worked directly with the founding team across a remote engagement, iterating on scope and shipping features to an agreed timeline.

Freelance Backend Developer — Self-Employed

Jan 2021 – Aug 2025

Mumbai, India · Remote · Python, Django, DRF, Django Channels, PostgreSQL, Docker

- **Delivered 10 production systems** across real estate, call-centre operations, retail and e-commerce, and import/export — solo across the full lifecycle: requirements, development, deployment and maintenance.
- **Engineered role-based access systems and real-time features** using Django Channels and WebSockets, on PostgreSQL-backed CRMs and e-commerce platforms.
- **Deployed and maintained applications** on Azure, DigitalOcean, Render and Hostinger using Docker, including on-premise installs at client sites.
- **Served clients in two countries** (India and Germany), maintaining codebases that client teams depended on daily.

SELECTED PROJECTS

Customer Shopping Behaviour Analysis — End-to-End Data Pipeline — Python, pandas, SQLAlchemy, PostgreSQL, Power BI

- Built an ETL pipeline over a 3,900-row retail dataset: cleaned in pandas with category-level median imputation, loaded into PostgreSQL via SQLAlchemy, then answered ten business questions in analytical SQL using CTEs and RANK() OVER (PARTITION BY category).
- Derived age-quartile and purchase-cadence fields, added a data-quality framework for schema integrity, completeness and anomaly checks, and published findings in an interactive Power BI dashboard.

InitCore CRM — Call Centre Platform — Django 5, PostgreSQL, Django Channels, Daphne, pandas

- Designed a 17-model Django application for an outbound call-centre desk — lead distribution, call outcomes, payments, GST invoice generation as PDFs, attendance and live break monitoring over WebSockets.
- Implemented CSV/XLSX lead import with an interactive column-mapping step and de-duplication against existing records, plus full audit history and role-scoped queriesets so one set of views serves admins, team leaders and agents.

FlaskSearch — REST Search API — Flask, Elasticsearch, Docker

- Built a REST search service running multi_match queries across title, author and character fields against an Elasticsearch index, with endpoints for listing, searching and indexing documents and a Dockerised local setup.

ADDITIONAL CLIENT DELIVERY

- **key2yourhome.net** — property listing marketplace for a MahaRERA-registered channel partner on Django 5.2 and PostgreSQL, with OAuth authentication, a Grappelli admin interface and multi-parameter property search across Mumbai, Thane, Navi Mumbai and Pune.
- **InitCore Real Estate CRM** — Django platform for co-living property management, tracking bed-space inventory, tenancies and billing.
- **S.K Trading & Co.** — offline Electron billing application installed on site for an industrial valve supplier (quotations, proforma and GST tax invoices, delivery challans), plus a 224-product public catalogue site.
- **tajmetals.in and Logxivia** — catalogue and enquiry sites for a Mumbai steel supplier and a Cologne-based Europe–India trade consultancy, the latter built GDPR-aware with essential-only cookies and no tracking.
- **ICTMT 2025 and SUSTECH 2025** — websites for two international conferences at Saraswati College of Engineering, built content-as-data so organisers update deadlines, fees and committees without touching component markup.

EDUCATION

B.E., Computer Science and Engineering (Data Science)

2023 – Jul 2026

Saraswati College of Engineering, University of Mumbai · Navi Mumbai, India

CGPI 8.19 / 10 (73.73%) · All semesters cleared on first attempt · Direct second-year entry on the strength of the diploma

Relevant coursework: Big Data Analytics, Machine Learning, Deep Learning, Natural Language Processing, Data Warehousing & Mining, Data Analytics & Visualisation, Distributed Computing, Cloud Computing, DBMS.

Diploma in Computer Engineering

2021 – Jun 2023

A. R. Kalsekar Polytechnic, MSBTE · New Panvel, India

First Class — 72.63% aggregate · No backlogs · Final semester 75.53% (First Class with Distinction)

HSC (Science), Maharashtra State Board — 87.50% (2021) · SSC — 81.20% (2019) · IELTS Academic — Band 6.5, CEFR B2 (June 2026)

ACHIEVEMENTS & CERTIFICATIONS

- **Smart India Hackathon 2025 — Grand Finalist (Team Leader).** National hackathon run by the Ministry of Education's Innovation Cell and AICTE; selected from over 50,000 competing teams and competed in the 36-hour Grand Finale at BPUT, Rourkela, 8–9 December 2025.
- **SCOE Avishkar 2025 — 2nd Place (Team Leader).** Project competition at Saraswati College of Engineering, University of Mumbai, April 2025.
- **Certifications:** Complete Data Analyst Bootcamp — Udemy (Feb 2026) · The Complete Full-Stack Web Development Bootcamp — Udemy (Feb 2026) · Excel Basics for Data Analysis and Introduction to Data Analytics — Coursera / IBM (Dec 2025 – Jan 2026) · Introduction to Data Science — Great Learning (Oct 2025).